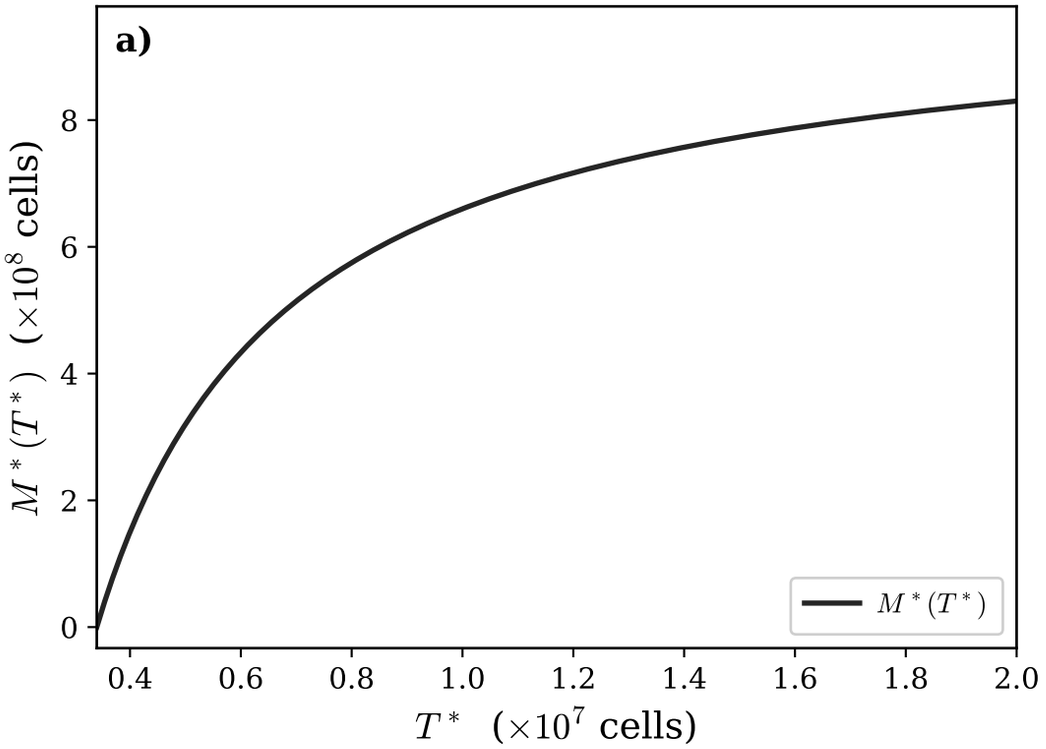
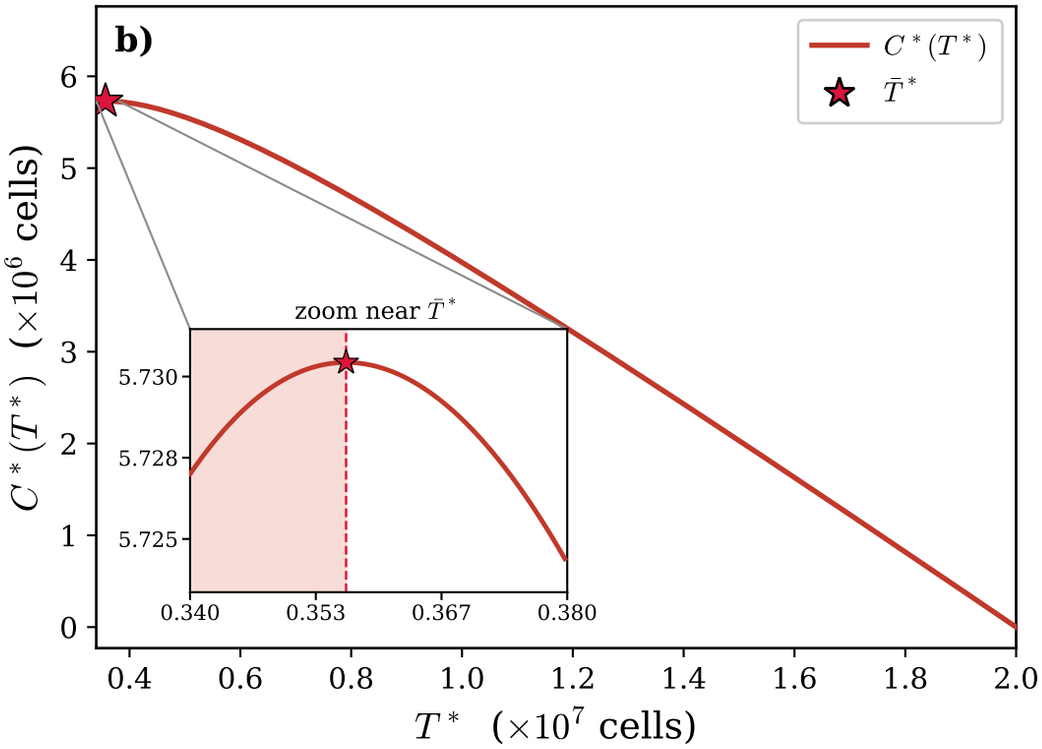
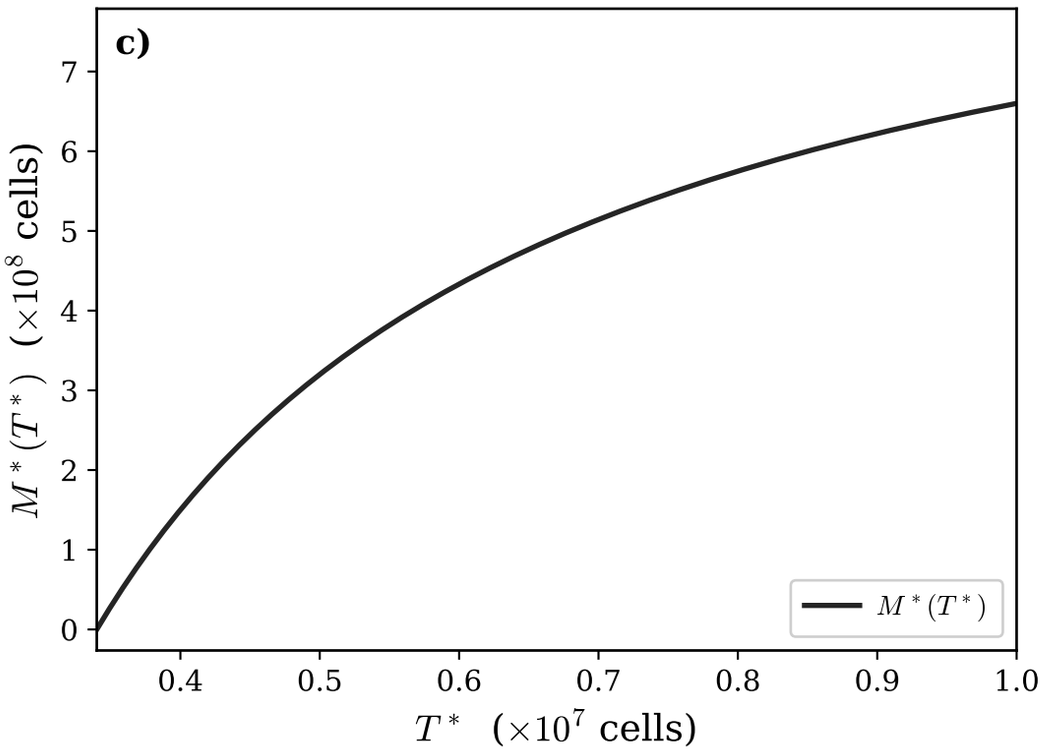
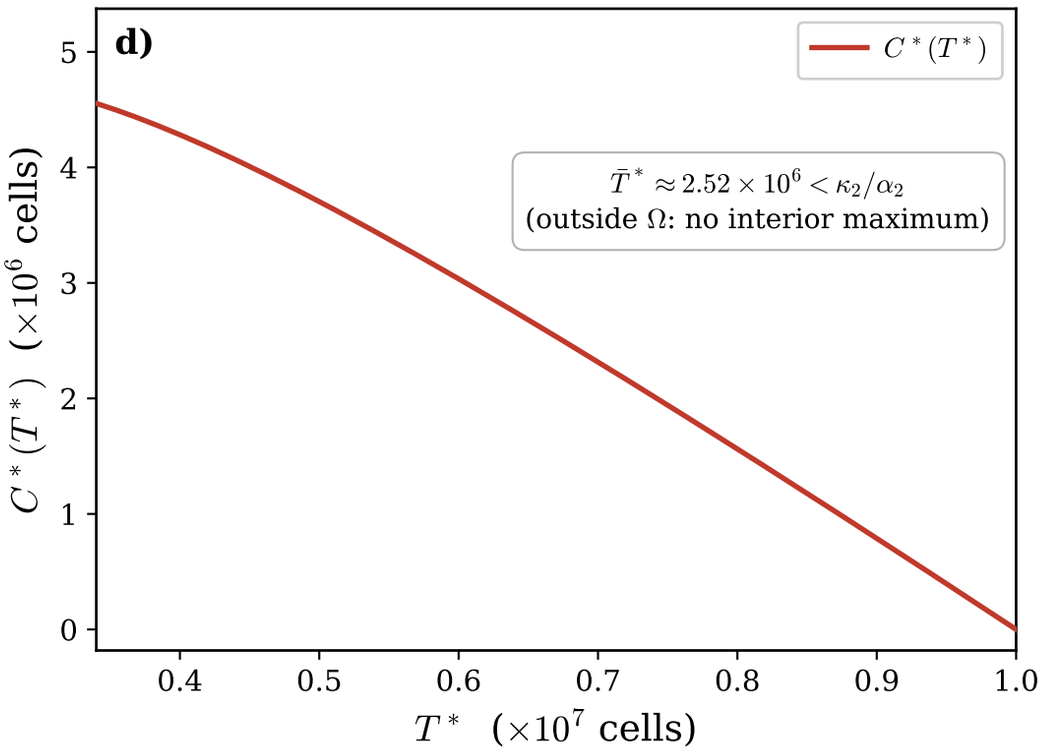
Case 1: $K_1 = 2 \times 10^7 > \bar{K}_1 \approx 1.82 \times 10^7$ cells





Case 2: $K_1 = 1 \times 10^7 \leq \bar{K}_1 \approx 1.82 \times 10^7$ cells